Special Algebraic Identities Worksheet

Name: _____ **Date:** _____

Special Identities Reference

Identity Name	Formula
1. Perfect Square (Addition)	$(a + b)^2 = a^2 + 2ab + b^2$
2. Perfect Square (Subtraction)	$(a - b)^2 = a^2 - 2ab + b^2$
3. Difference of Two Squares	$(a + b)(a - b) = a^2 - b^2$

Part 1: Expand Simple Algebraic Expressions

Expand the following expressions using the special identities.

1. $(x + 4)^2$
2. $(y - 6)^2$
3. $(a + 9)(a - 9)$
4. $(2x + 3)^2$
5. $(5m - 2)^2$
6. $(3p + 4q)(3p - 4q)$
7. $(4k + 1)^2$
8. $(7 - 2y)^2$
9. $(10a + 3b)(10a - 3b)$
10. $(½ x + 4)^2$

Part 2: Factorise Algebraic Expressions

Factorise the following expressions completely using the special identities. Look for perfect square trinomials and differences of two squares.

11. $x^2 + 10x + 25$
12. $y^2 - 14y + 49$
13. $m^2 - 36$
14. $4a^2 + 12a + 9$
15. $16x^2 - 24x + 9$
16. $25p^2 - 64q^2$
17. $81 - b^2$
18. $9x^2 + 30xy + 25y^2$
19. $49m^2 - 144n^2$
20. $2x^2 - 18$ *(Hint: Factor out the greatest common factor first)*

Part 3: Apply Identities to Solve Basic Problems

Use the special identities to calculate the numerical values or solve the equations.

21. Evaluate 41^2 without a calculator by rewriting it as $(40 + 1)^2$.
22. Evaluate 98^2 without a calculator by rewriting it as $(100 - 2)^2$.
23. Evaluate 102 x 98 using the difference of two squares identity.
24. Solve for x: $(x + 3)^2 - (x - 3)^2 = 24$.
25. If $a + b = 7$ and $ab = 10$, find the value of $a^2 + b^2$. *(Hint: Expand $(a+b)^2$)*

Part 4: Explain the Steps

Provide clear, step-by-step written explanations for the following problems.

26. **Explain the steps** to expand $(3x - 5)^2$. Identify which special identity you are using and clearly state what your a and b values are before solving.
27. A student expanded an expression and wrote: $(x + 7)^2 = x^2 + 49$. **Identify the error** they made. Explain what term is missing and provide the correct expansion.

28. **Describe how to recognize** that $16x^2 - 40x + 25$ is a perfect square trinomial. What features of the first and last terms tip you off, and how do you verify the middle term?
29. **Write down the step-by-step process** to factorise $50x^2 - 8$. Explain why you cannot apply the difference of two squares identity immediately to the original expression.
30. **Explain how** the identity $(a+b)(a-b) = a^2 - b^2$ can be used to quickly calculate the difference between the squares of two large numbers, such as $85^2 - 15^2$. Show the calculation in your explanation.